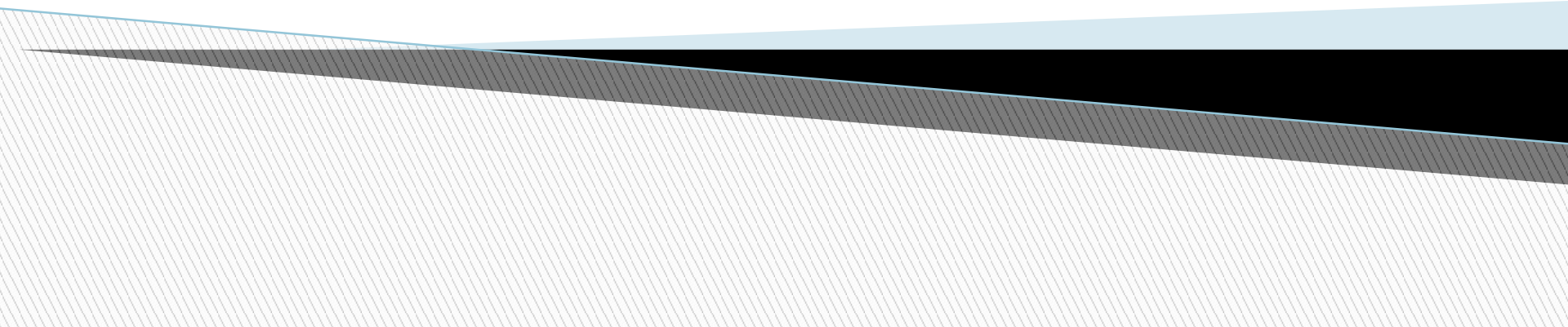


PSOT

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Special random variables



Types of Distributions

☐ Continuous Distribution

- ☐ Uniform
- ☐ Gaussian
- ☐ Exponential

☐ Discrete Distribution

- ☐ Bernoulli
- ☐ Binomial
- ☐ Poisson

Uniform Distribution

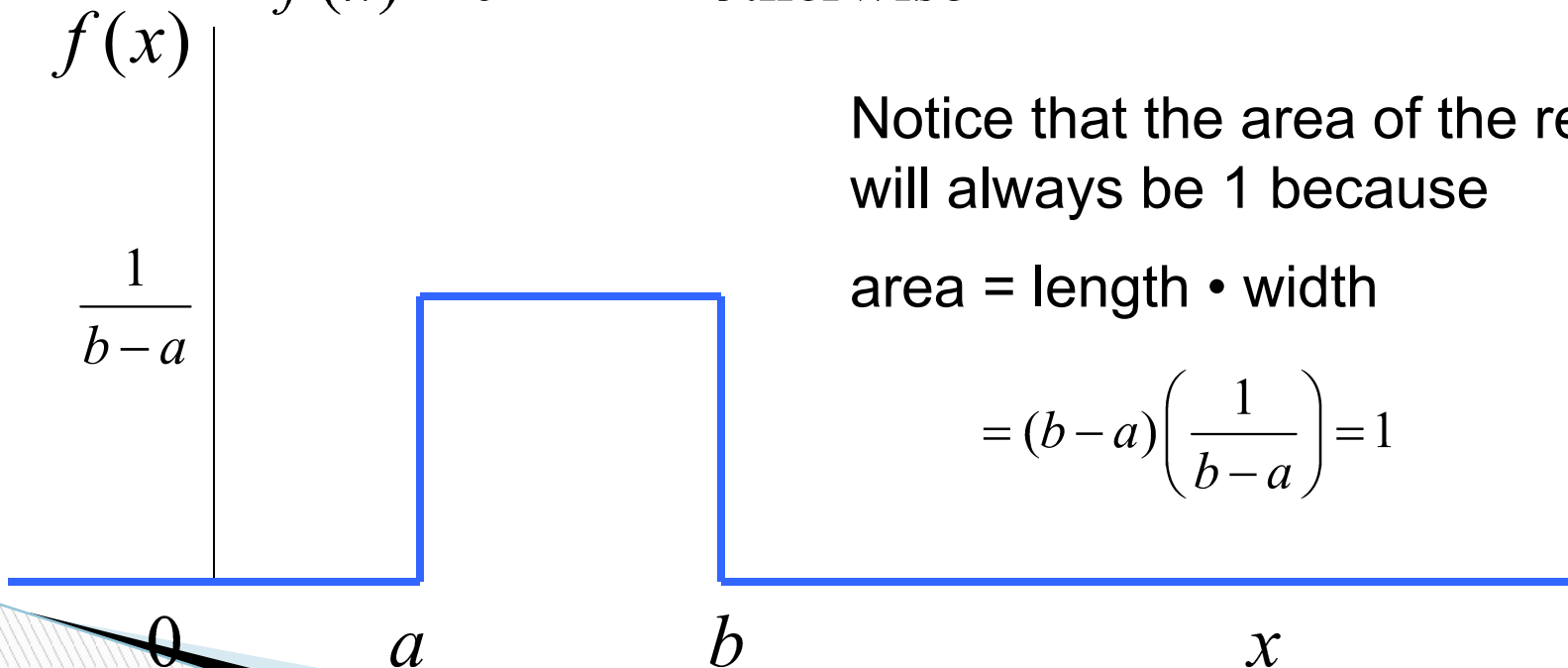
- There are both **discrete** uniform distributions and **continuous** uniform distributions

Uniform Distribution

The continuous uniform distribution has the following form:

$$f(x) = \frac{1}{b-a} \quad \text{if } a \leq x \leq b$$

$$f(x) = 0 \quad \text{otherwise}$$



Notice that the area of the rectangle will always be 1 because

area = length • width

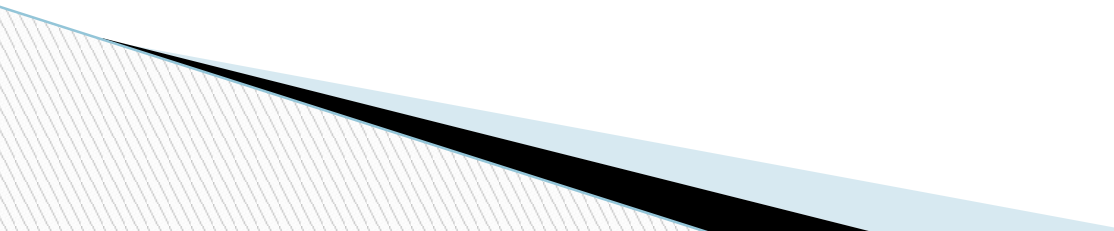
$$= (b-a) \left(\frac{1}{b-a} \right) = 1$$

Mean, variance, and standard deviation of the continuous uniform distribution

□ mean:
$$\mu = \frac{a + b}{2}$$

variance:
$$\sigma^2 = \frac{(b - a)^2}{12}$$

standard deviation:
$$\sigma = \sqrt{\frac{(b - a)^2}{12}}$$



The most famous distribution is the *Normal* or *Gaussian* distribution.

□ Its probability density function (pdf) is

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}[(x-\mu)/\sigma]^2} \quad \text{for } -\infty < x < \infty$$

μ is the mean of the distribution, σ is the standard deviation, $e \approx 2.718$, and $\pi \approx 3.14$.

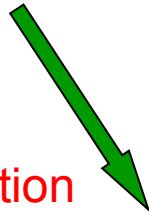
It is sometimes denoted $N(\mu, \sigma^2)$, which means the normal distribution with a mean of μ and a variance of σ^2 .

If you had the same mean but different standard deviations, it would look like this:

small standard deviation



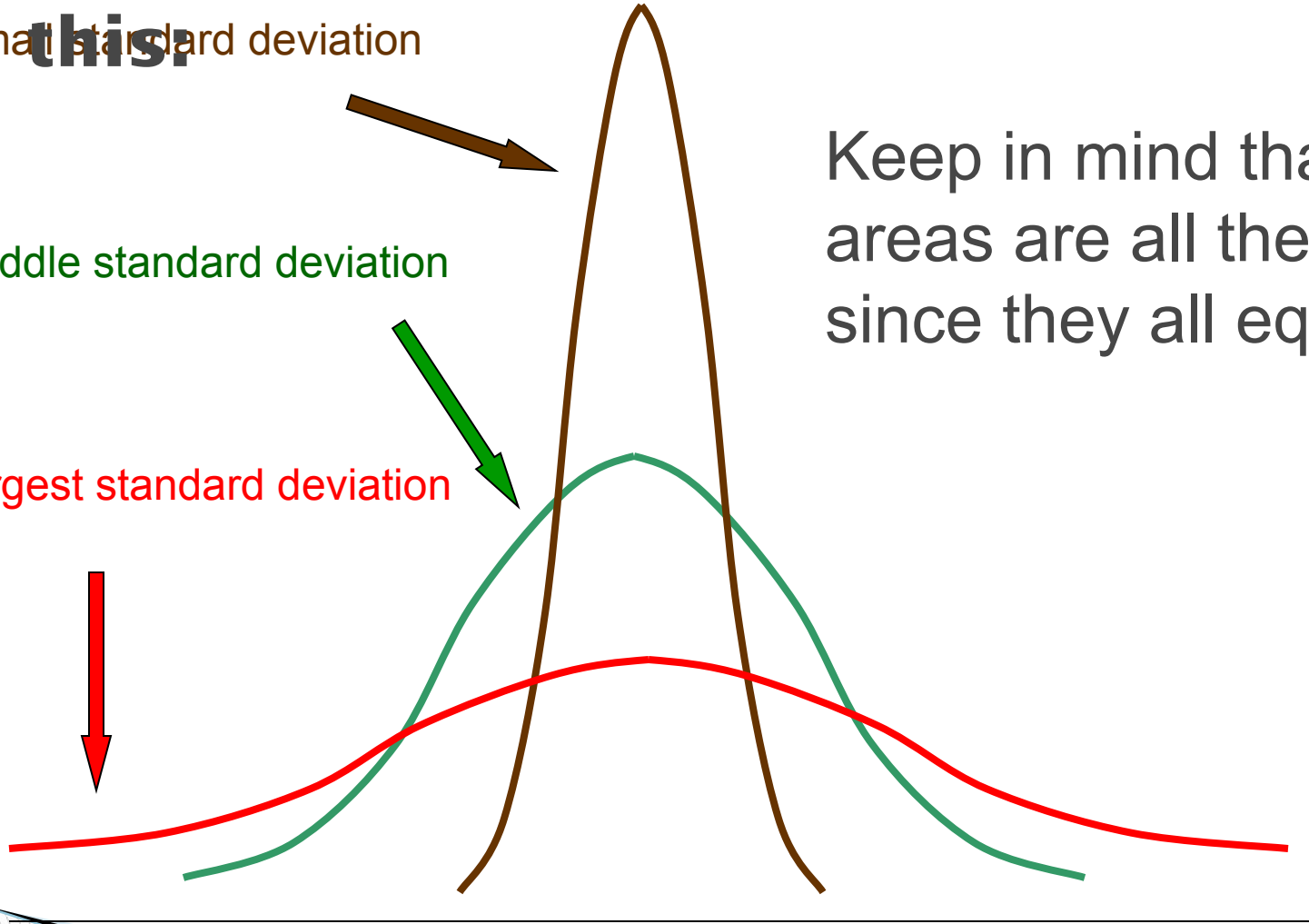
middle standard deviation



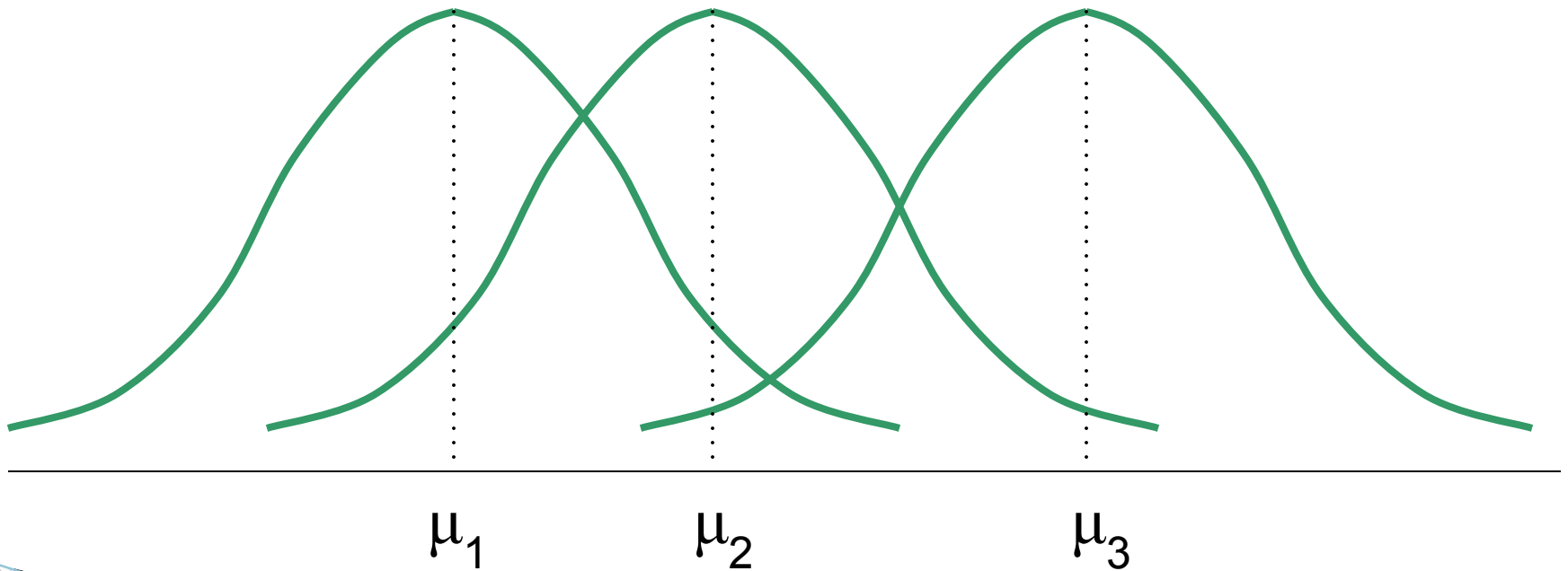
largest standard deviation

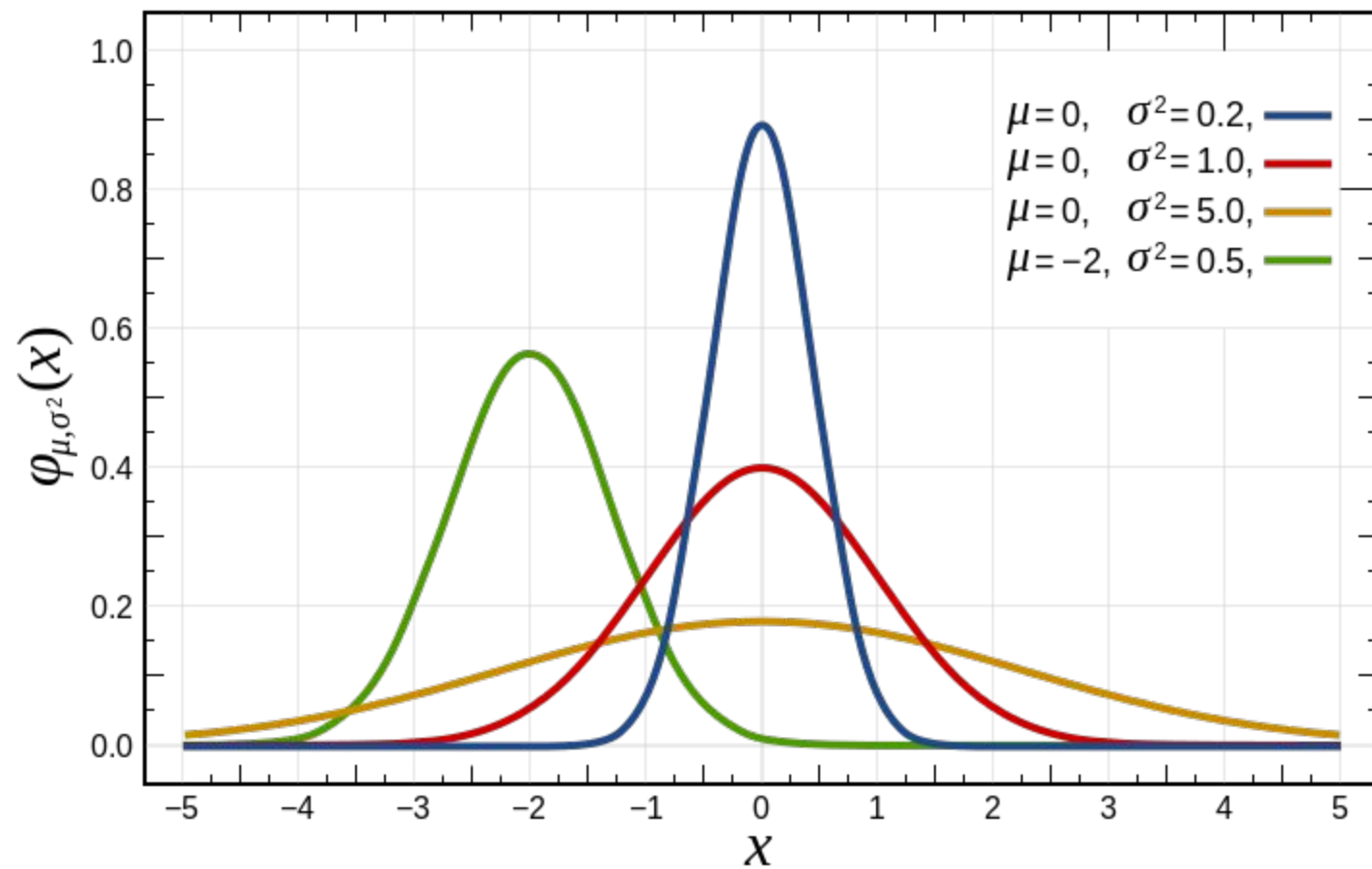


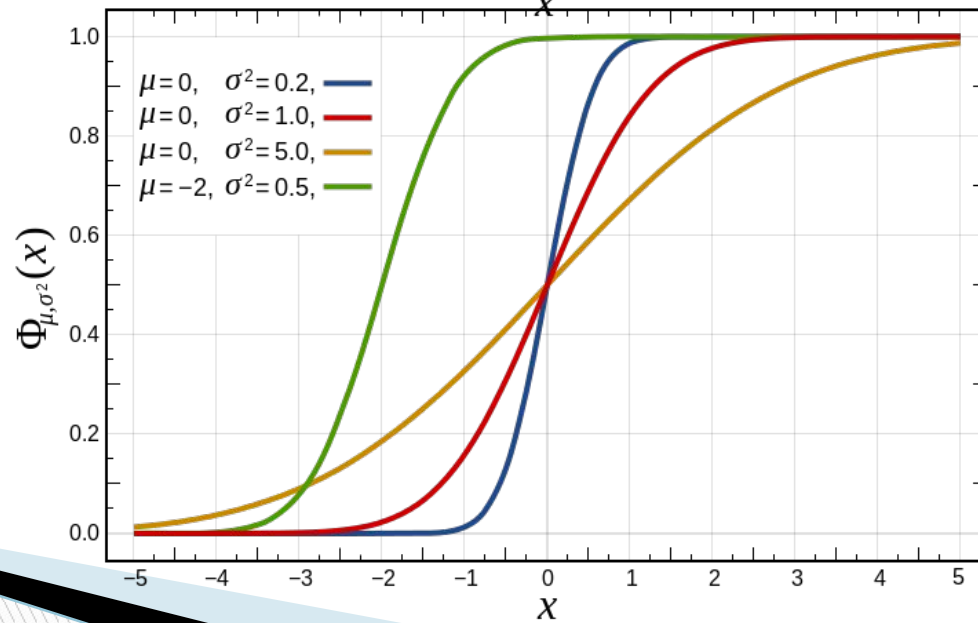
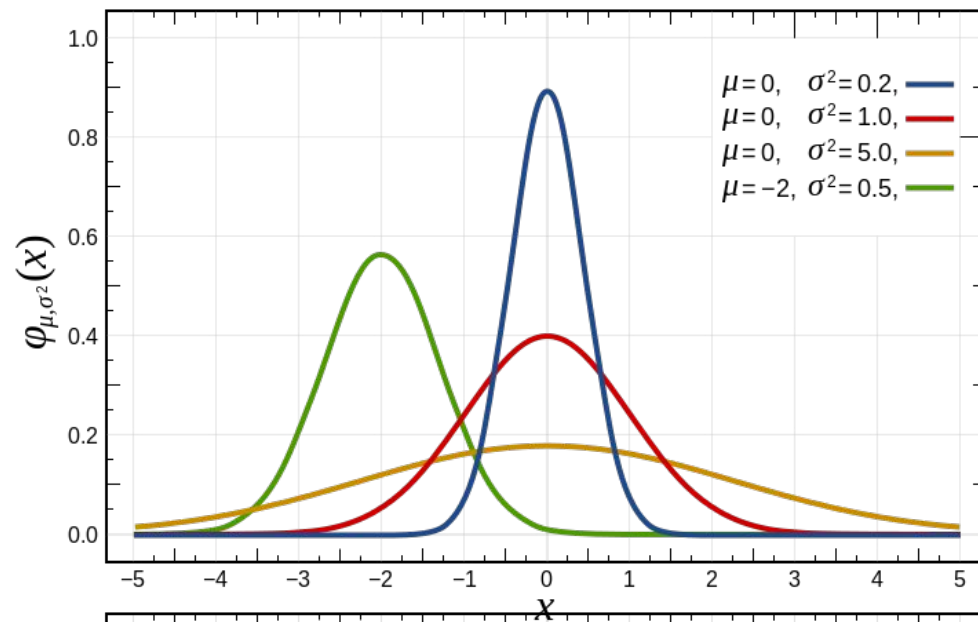
Keep in mind that the areas are all the same, since they all equal 1.



If you have three normal distributions with the same standard deviation (same spread), but different means (different averages), they would look like this:



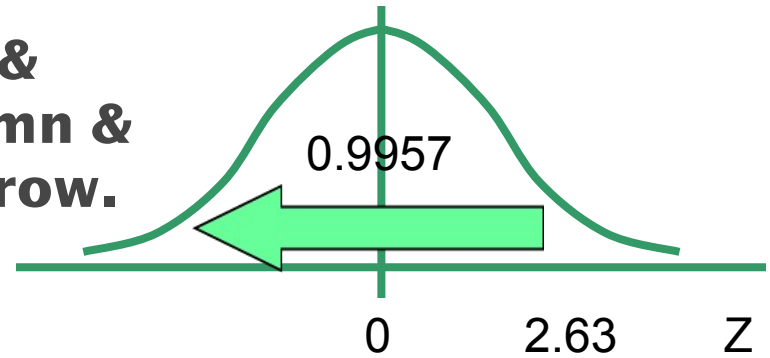




Normal or Gaussian **distribution**

- If X is normally distributed, then $(X-\mu)/\sigma$ will be standard normal, $N(0,1)$, normal with mean 0 and variance 1.

Z table: You get the integer part & the 1st decimal from the left column & the second decimal from the top row.

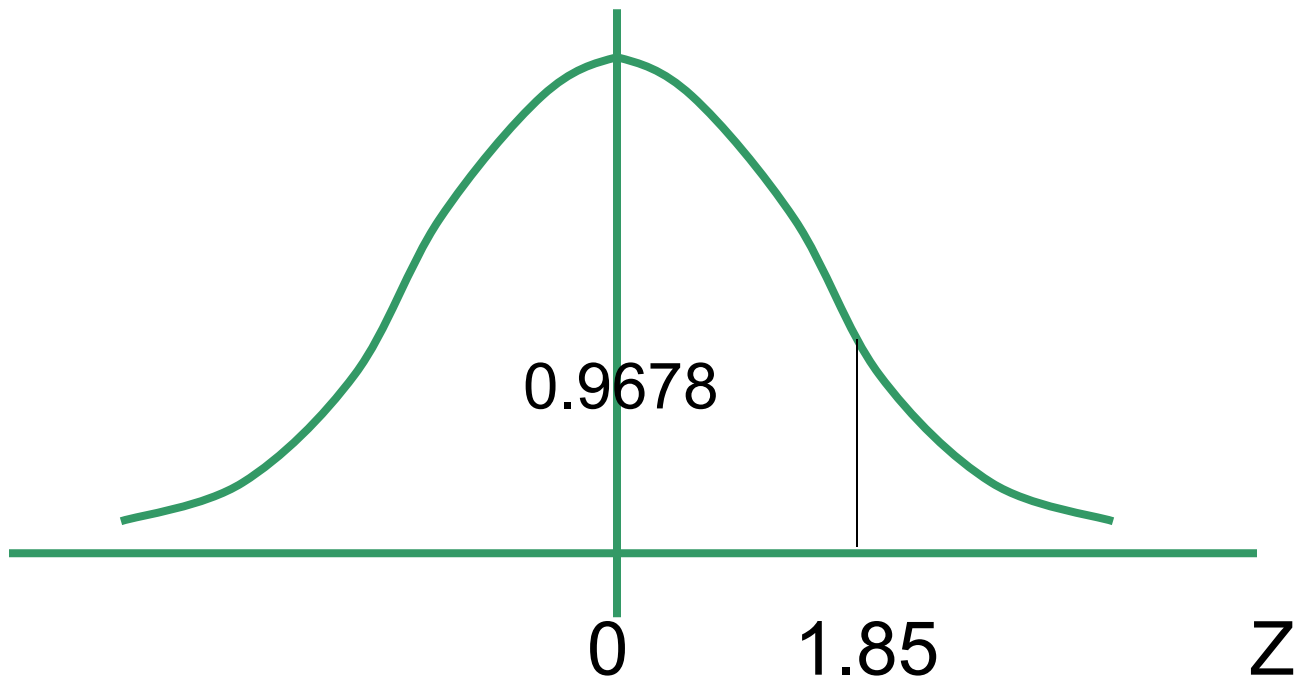


Example: $\Pr(Z \leq 1.23) = 0.8907$

[illegible]

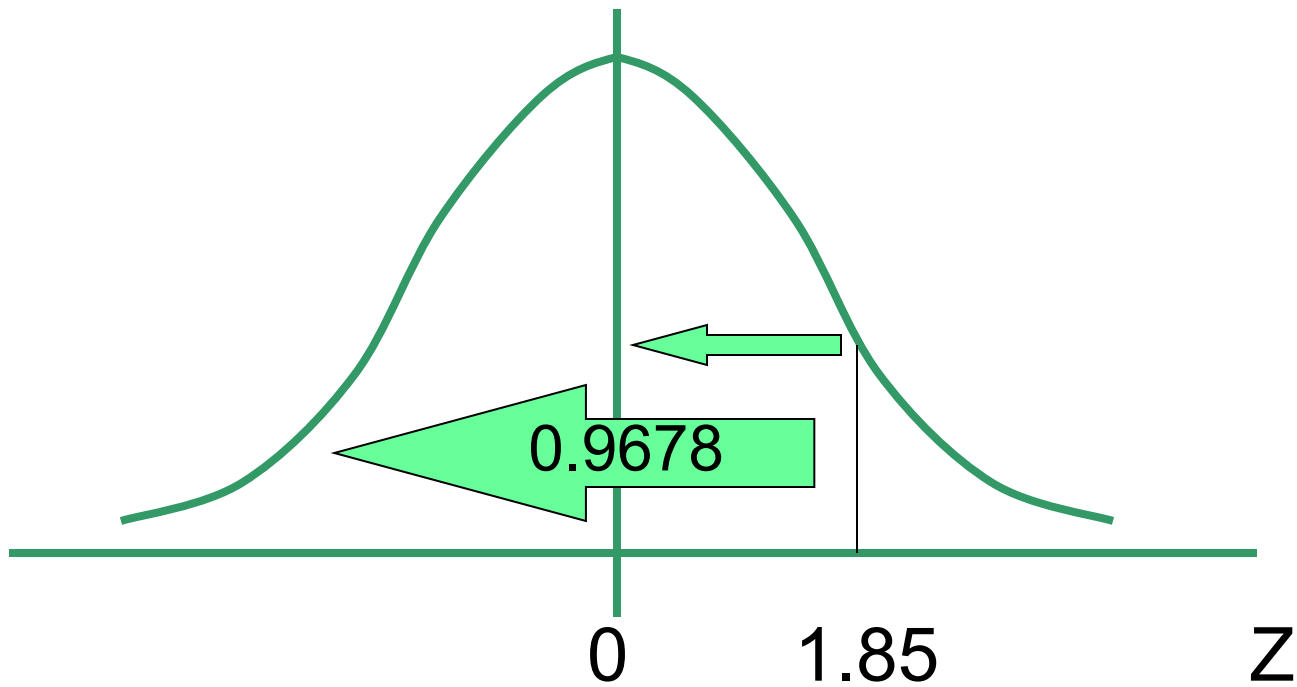
Example

$$\square \Pr(Z < 1.85) = 0.9678$$



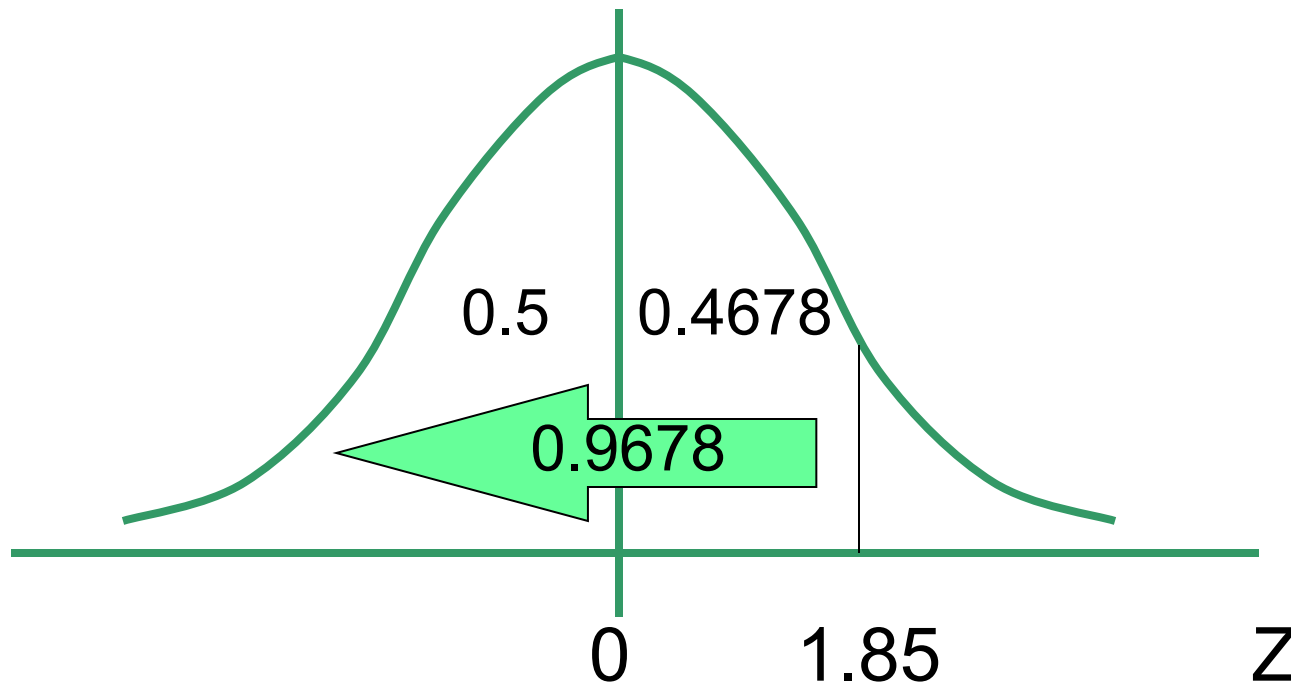
Example

$$\square \Pr(0 < Z < 1.85)$$



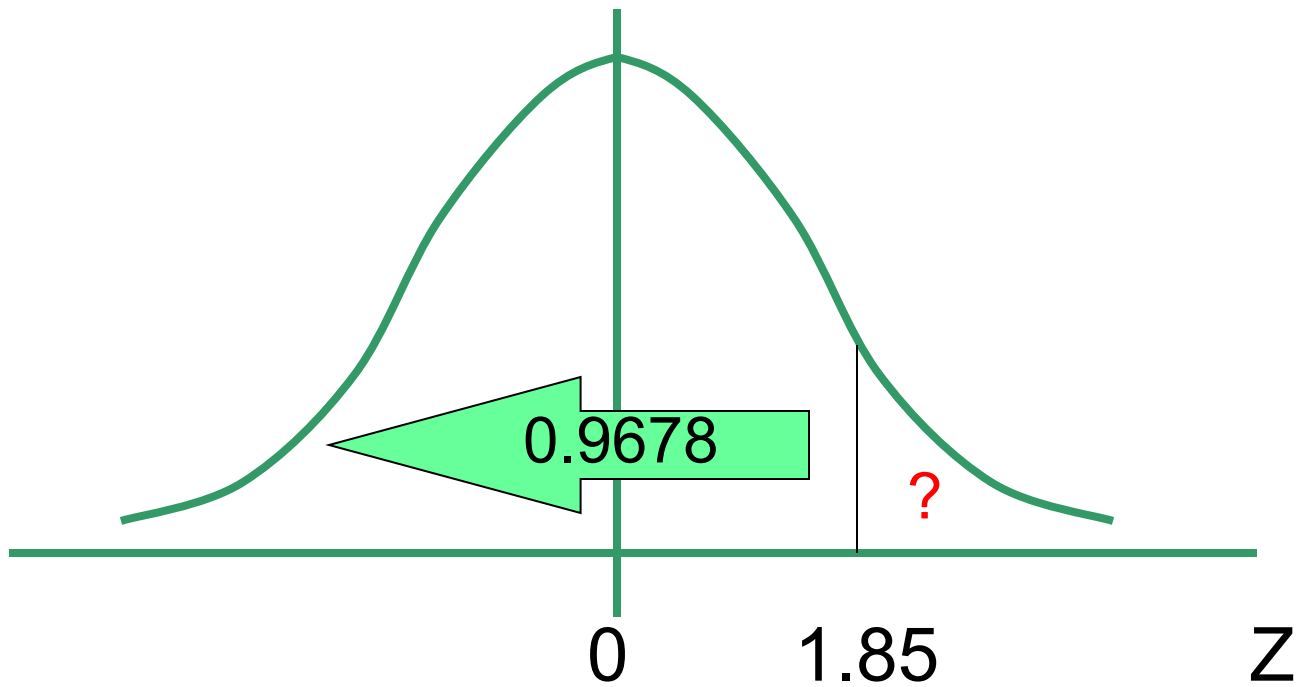
Example

$$\square \Pr(0 < Z < 1.85) = 0.9678 - 0.5 = 0.4678$$



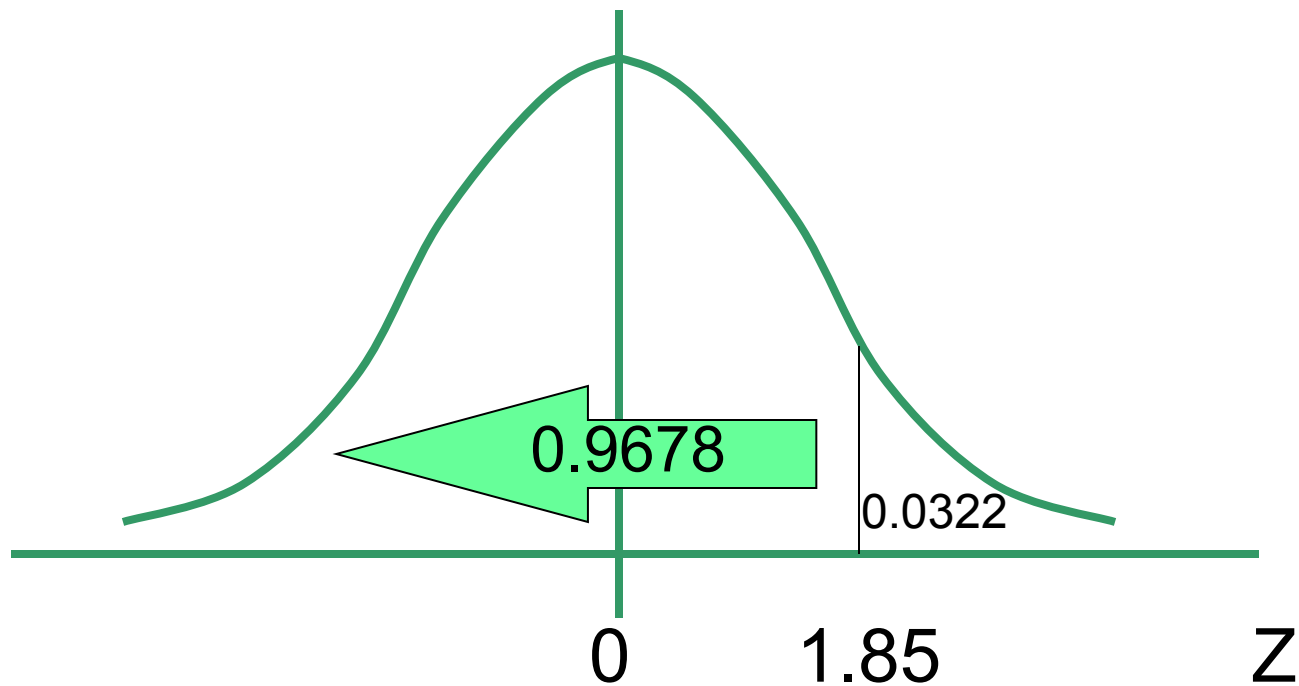
Example

□ $\Pr(Z > 1.85)$



Example

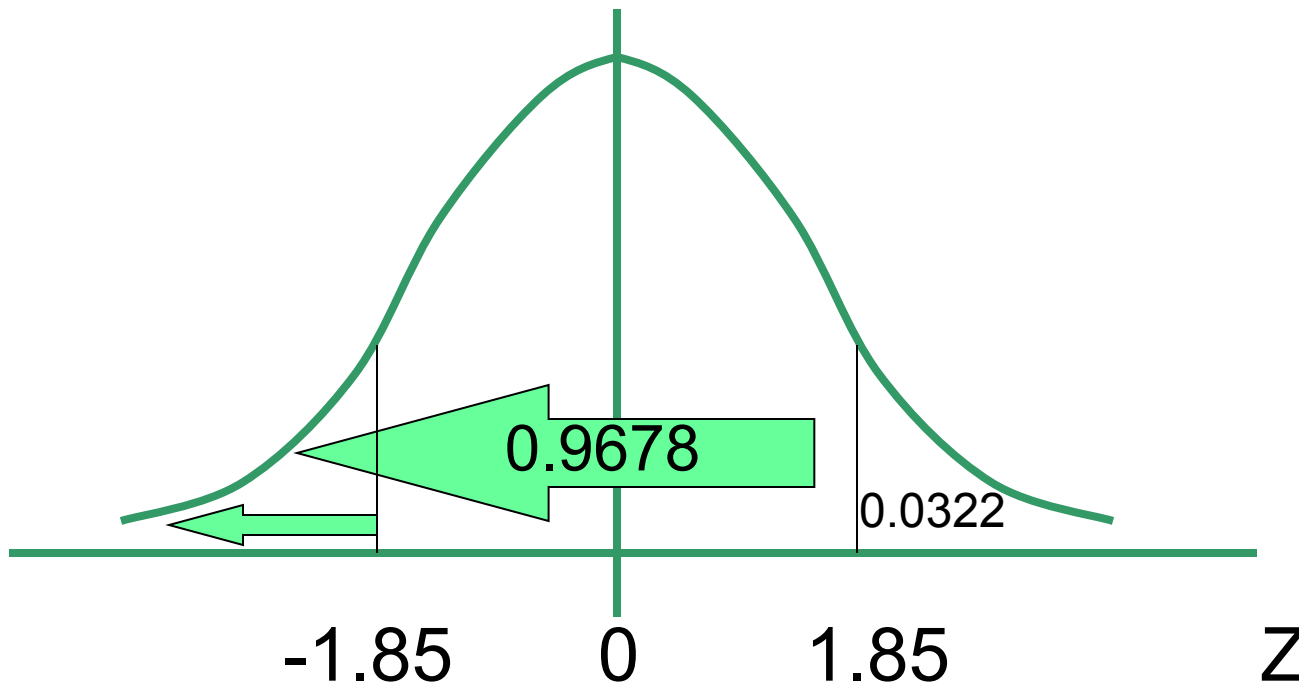
□ $\Pr(Z > 1.85) = 1 - 0.9678 = 0.0322$



Example

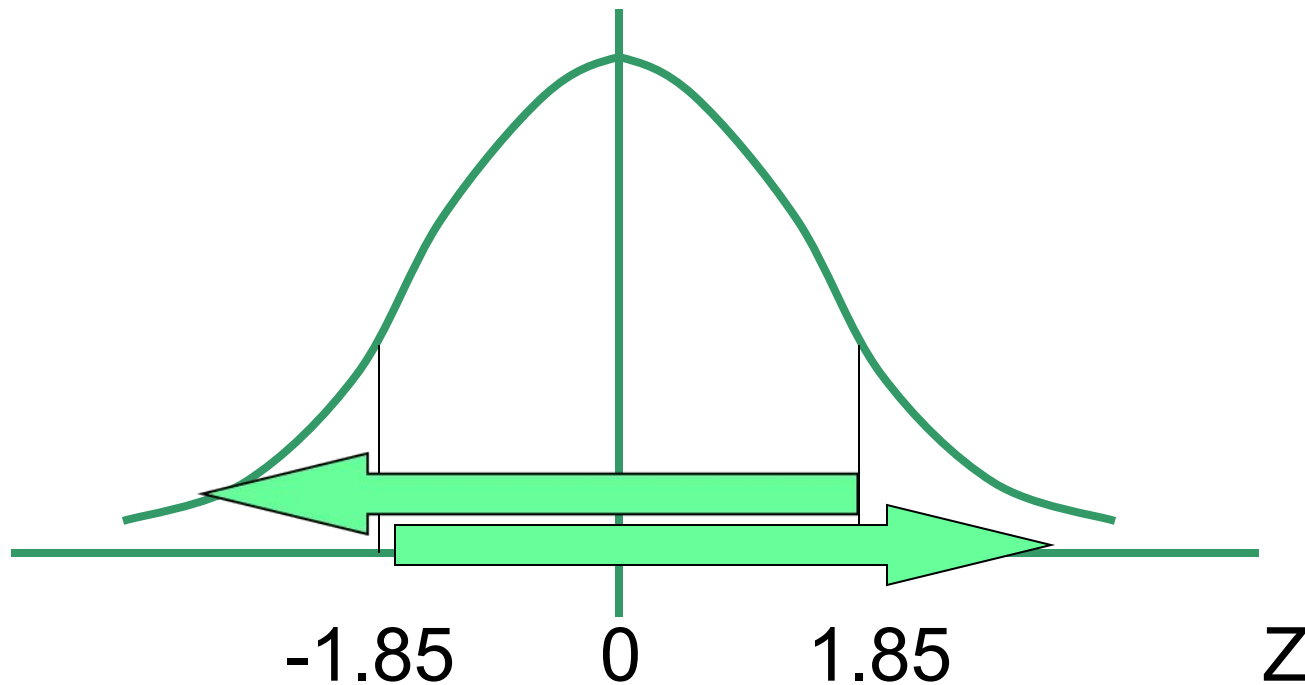
$$\square \Pr(Z < -1.85)$$

$$\Pr(Z < -1.85) = \Pr(Z > 1.85) = 0.0322$$



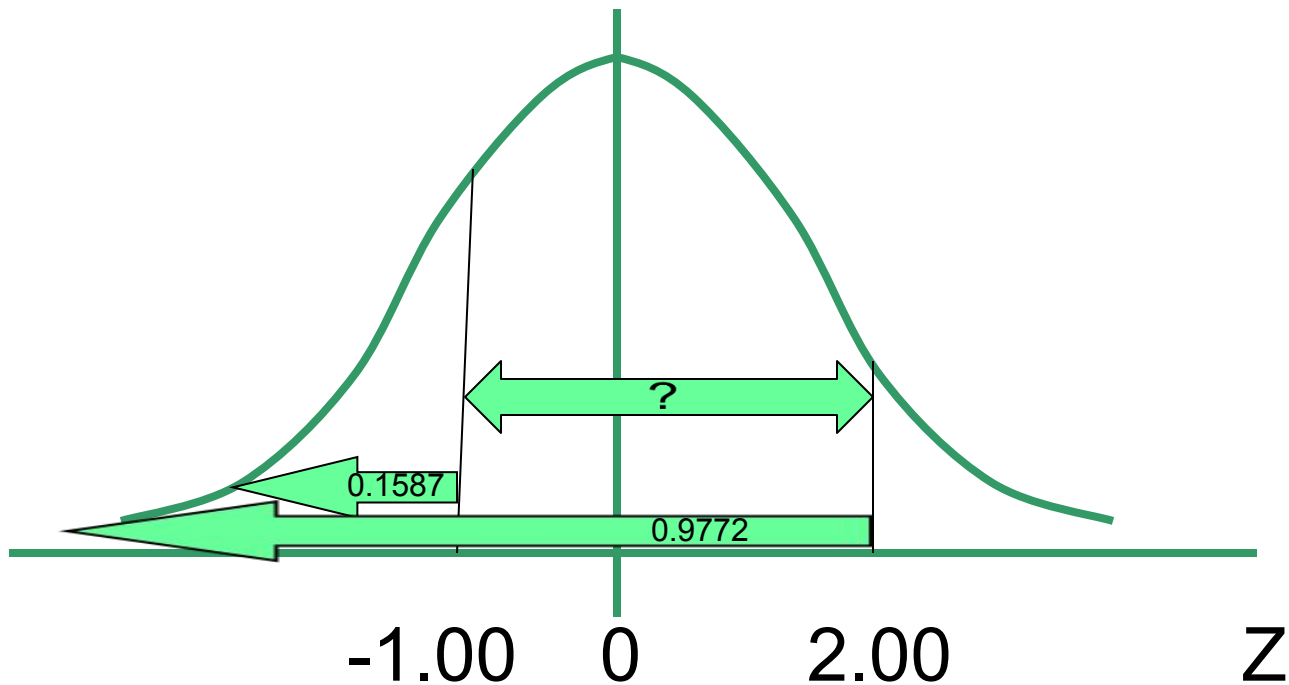
Example

$$\square \Pr(Z > -1.85) = \Pr(Z < 1.85) = 0.9678$$



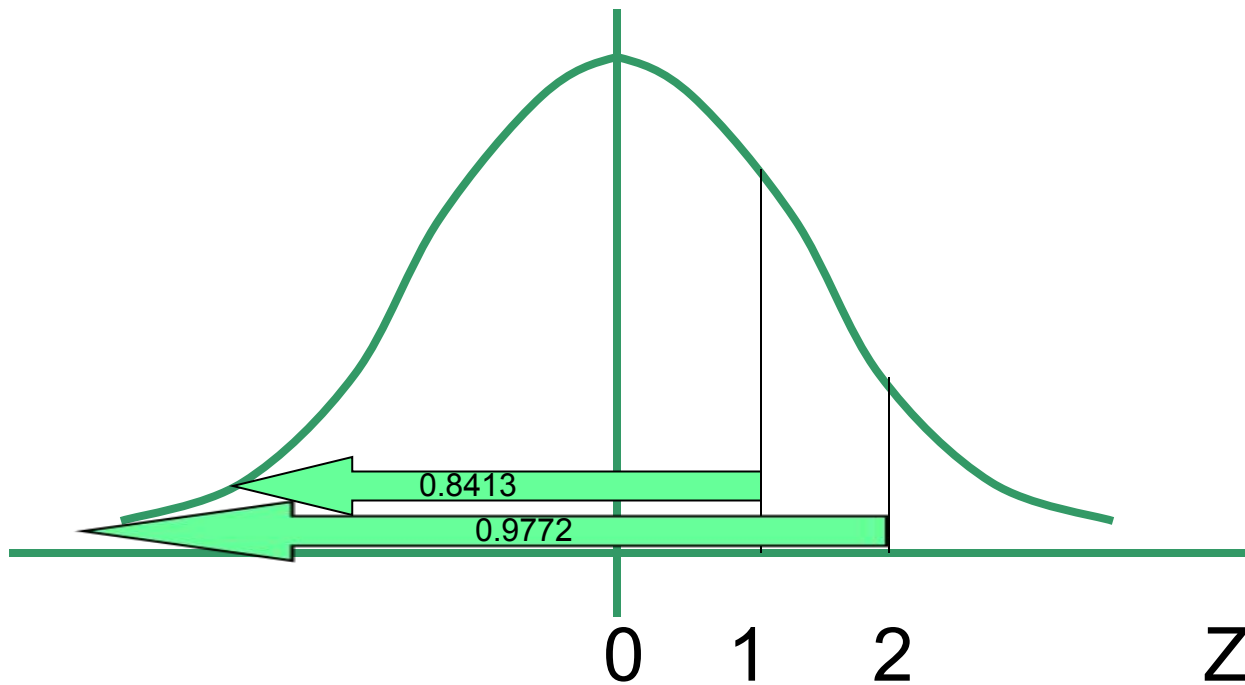
Example

$$\square \Pr(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185$$



Example

$$\square \Pr(1 < Z < 2) = 0.9772 - 0.8413 = 0.1359$$



Example:

If X is $N(2, 9)$, determine $\Pr(X \leq 5)$.

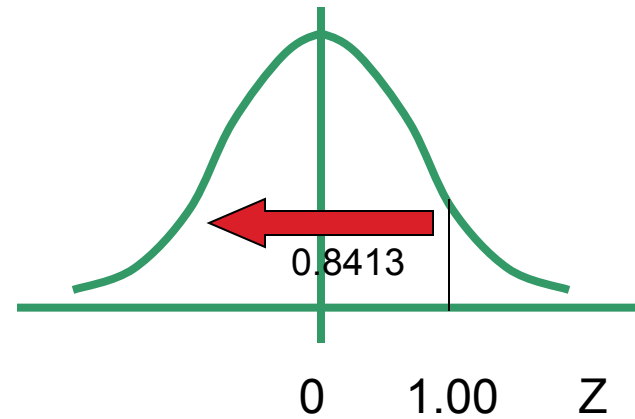
$$\Pr(X \leq 5)$$

$$= \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{5 - \mu}{\sigma}\right)$$

$$= \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{5 - 2}{3}\right)$$

$$= \Pr(Z \leq 1)$$

$$= 0.8413$$



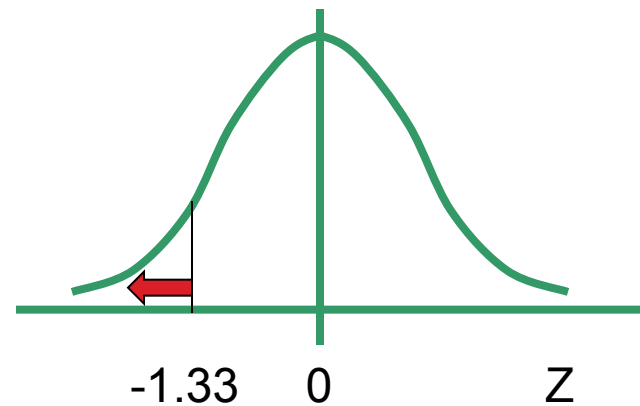
Example: Suppose that women's heights are normally distributed with mean 64 inches & standard deviation 3 inches. What is the probability that a randomly selected woman is under five feet tall?

$$\Pr(X \leq 60)$$

$$= \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{60 - \mu}{\sigma}\right)$$

$$= \Pr\left(\frac{X - \mu}{\sigma} \leq \frac{60 - 64}{3}\right)$$

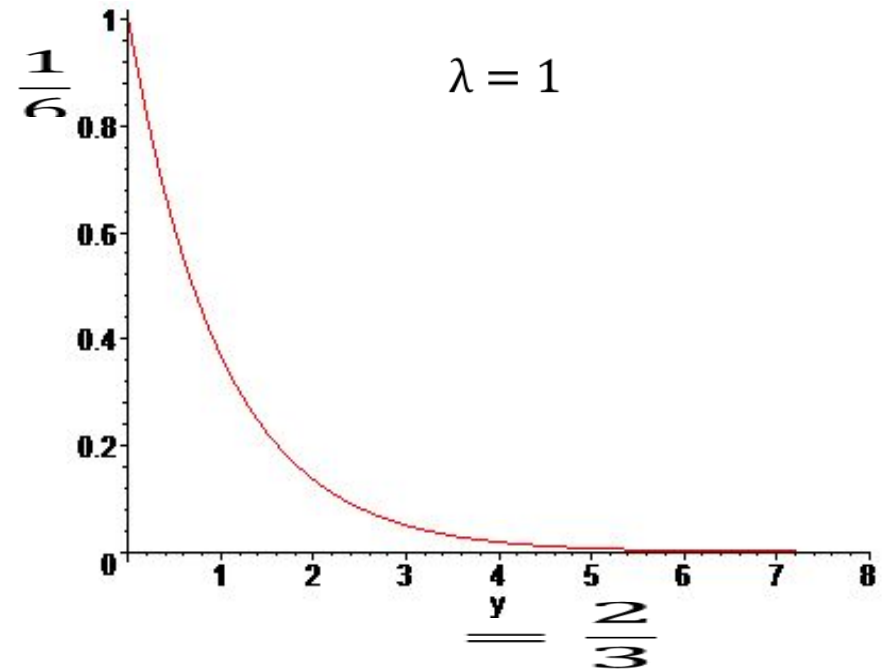
$$= \Pr(Z \leq -1.33) = 0.0918$$



Exponential distribution

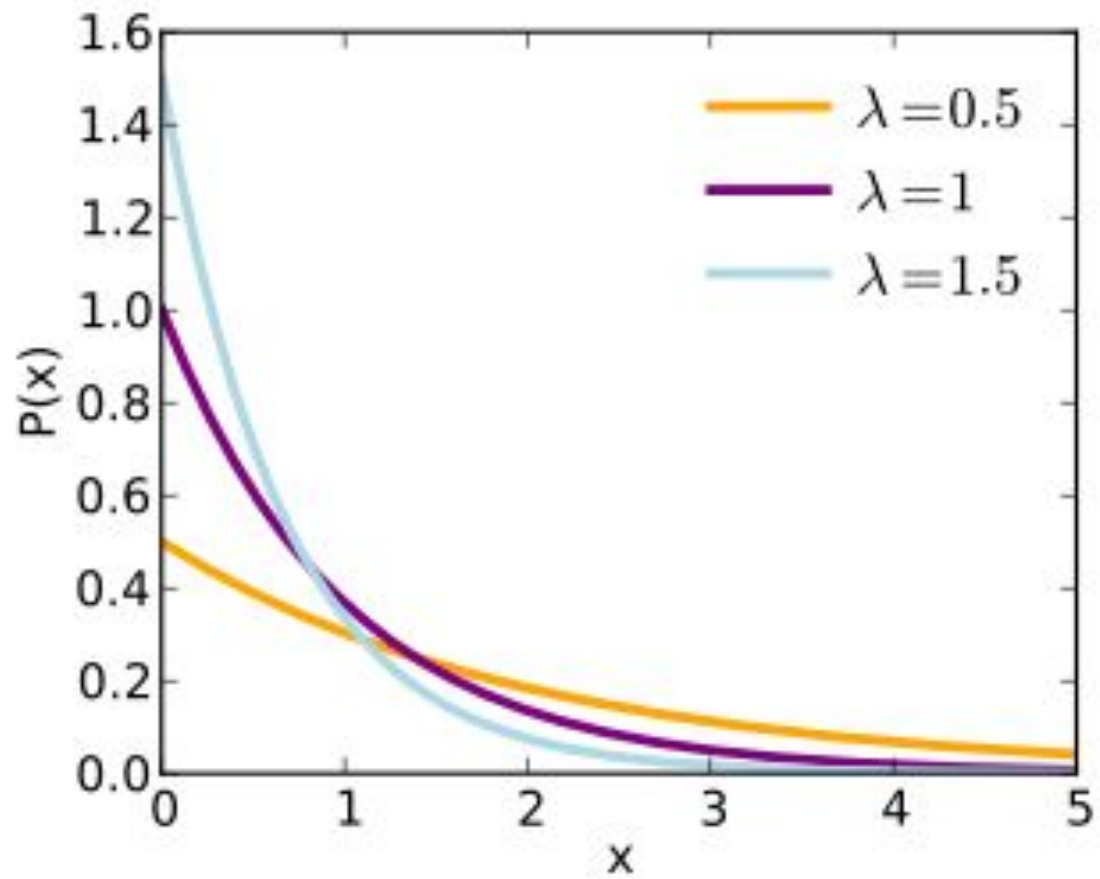
The continuous random variable Y has the Exponential distribution, with *constant* rate parameter λ if:

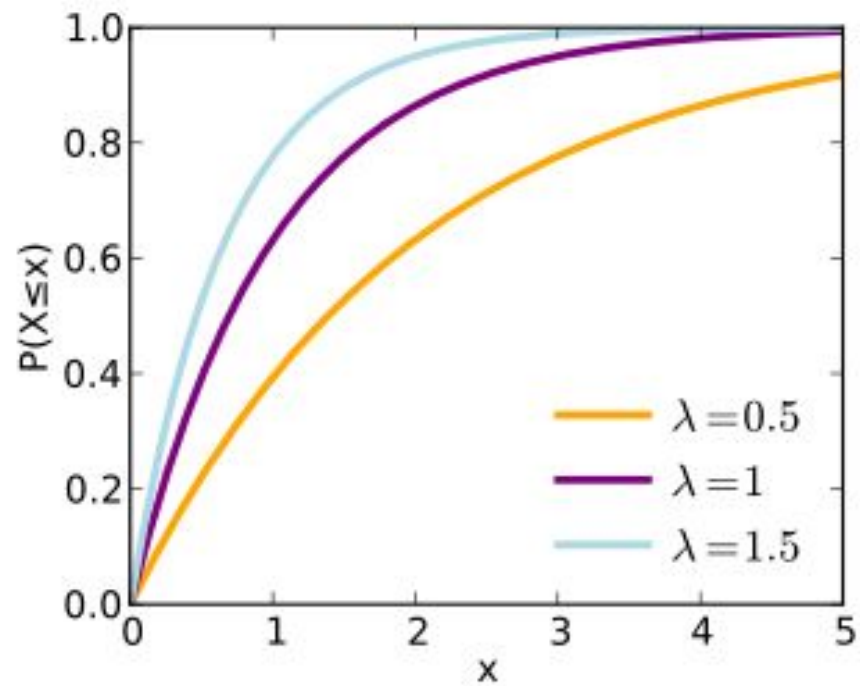
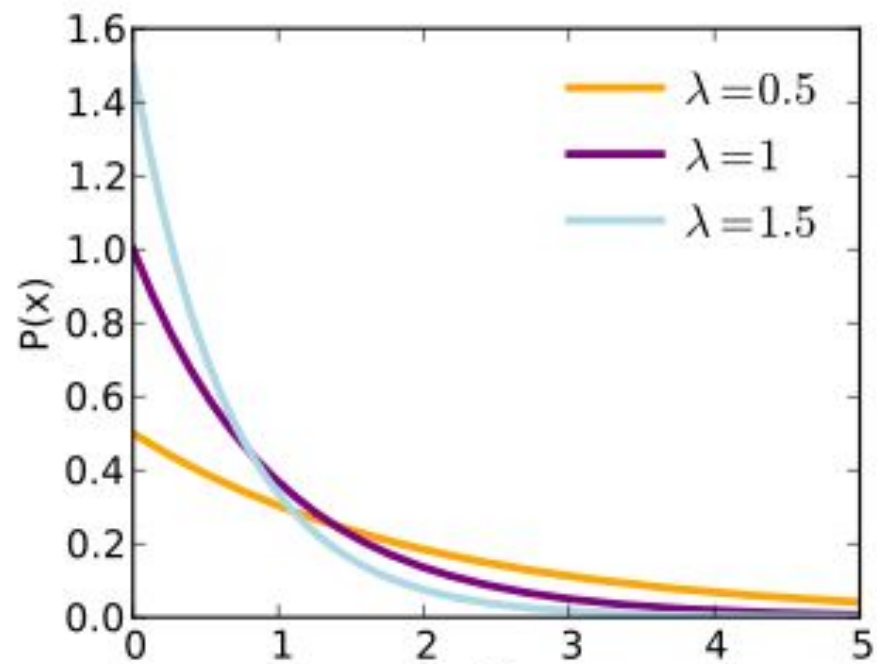
$$f(y) = \begin{cases} \lambda e^{-\lambda y}, & y > 0 \\ 0, & y < 0 \end{cases}$$



Occurrence

- 1) Time until the failure of a part.
- 2) Separation between randomly happening events
 - Assuming the probability of the events is constant in time: $\lambda = \text{const}$



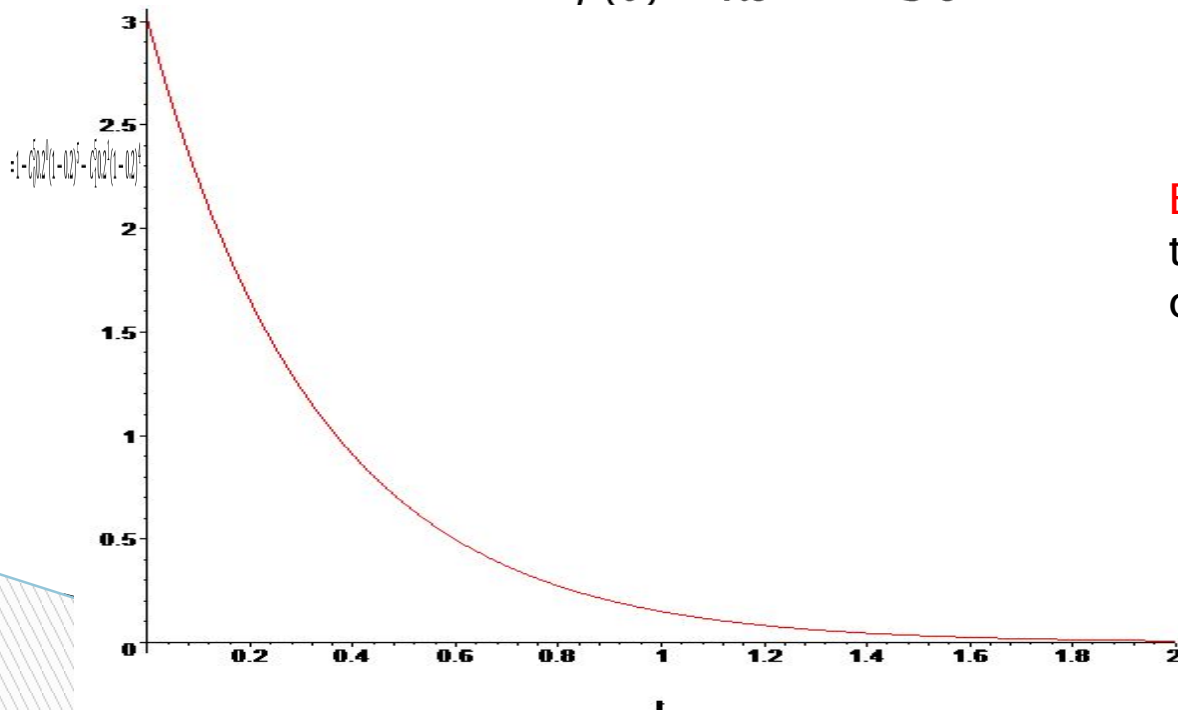


Example

On average lightening kills three people each year in the UK, $\lambda = 3$. So the rate is $\lambda = 3/\text{year}$.

Assuming strikes occur randomly at any time during the year so λ is constant, time from today until the next fatality has pdf (using t in years)

$$f(t) = \lambda e^{-\lambda t} = 3 e^{-3t}$$



E.g. Probability the time till the next death is less than one year?

$$= 1 - 0.32768 - 0.4096 \approx 0.263.$$

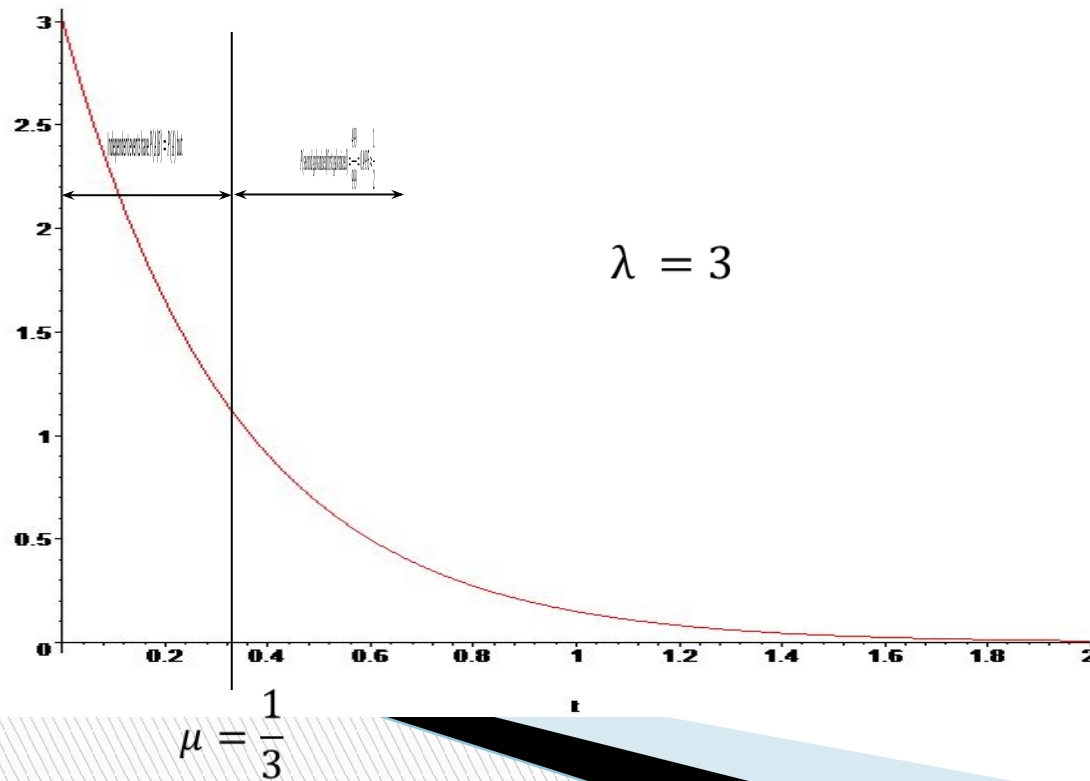
$$= \left[\frac{3e^{-3t}}{-3} \right]_0^1$$

Reminder: $P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!}$

Mean and variance of exponential distribution

$$\mu = \int_{-\infty}^{\infty} y f(y) dy = \int_0^{\infty} y \lambda e^{-\lambda y} dy = [-y e^{-\lambda y}]_0^{\infty} + \int_0^{\infty} e^{-\lambda y} dy = \left[-\frac{e^{-\lambda y}}{\lambda} \right]_0^{\infty} = \frac{1}{\lambda}$$

$$\begin{aligned} \sigma^2 &= \int_{-\infty}^{\infty} y^2 f(y) dy - \mu^2 = \int_0^{\infty} y^2 \lambda e^{-\lambda y} dy - \frac{1}{\lambda^2} \\ &= [-y^2 e^{-\lambda y}]_0^{\infty} + 2 \int_0^{\infty} y e^{-\lambda y} dy - \frac{1}{\lambda^2} = 0 + 2 \frac{\mu}{\lambda} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2} \end{aligned}$$



The time till failure of an electronic component has an Exponential distribution and it is known that 10% of components have failed by 1000 hours.

(a) What is the probability that a component is still working after 5000 hours?

(b) Find the mean and standard deviation of the time till failure.

In two hours mean number is $\lambda = 2 \times 3 = 6$.

Hence the variance for the sum of n independent trials is, by the rule for summing the variances of independent variables, $\sigma^2 = \sum_{i=1}^n \sigma_i^2 = np(1-p)$.

The variance in a single trial is:

$$\sigma_1^2 = \langle X^2 \rangle - \langle X \rangle^2 = 1^2 \times p + 0^2 \times (1-p) - p^2 = p(1-p)$$

$$P(X = k = 5) = \frac{e^{-\lambda} \lambda^k}{k!} = \frac{e^{-6} 6^5}{5!}$$

Mean and variance of a binomial distribution

$$\mu = np$$

$$\sigma^2 = np(1-p)$$

$$P(Y \leq 1000) = 0.1 \Rightarrow$$

If $X \sim \text{Poisson}$ with mean λ , then

$$\mu = E(X) = \lambda$$

$$\sigma^2 = \text{var}(X) = \lambda$$

Let k = number of messages in t seconds, so that

$$k \sim \text{Poisson, with } \lambda = 1.2 \times t = 1.2t$$

$$\text{Want } P(\text{At least one message}) = P(k \geq 1) = 1 - P(k = 0) \leq 0.05$$

$$\Rightarrow 1 - e^{-1.2t} \leq 0.05$$

$$\Rightarrow e^{-1.2t} \geq 0.95$$

$$\Rightarrow -1.2t \geq \ln(0.95) = -0.05129$$

$$\boxed{\Rightarrow t \leq 0.043 \text{ seconds}} \quad \Rightarrow 1 - e^{-1.2t} \leq 0.05$$

(b) Find the mean and standard deviation of the time till failure.

Answer:
r:

$$\Rightarrow -e^{-1.2t} \leq 0.05 - 1$$

$$P(k = 0) = \frac{e^{-\lambda} \lambda^k}{k!} = \frac{e^{-1.2t} (1.2t)^0}{0!} = e^{-1.2t}$$

Answer:

Times of arrivals form a Poisson process, rate $\nu = 1.2/\text{sec}$.

<i>X</i>	<i>X Measures</i>	<i>f(x)</i>	<i>Values of X</i>	<i>E(x)</i>	<i>V(x)</i>
Continuous uniform	Outcomes with equal density (continuous)	$\frac{1}{b-a}$	$a \leq x \leq b$	$\frac{b+a}{2}$	$\frac{(b-a)^2}{12}$
Exponential	Time between events; time until an event	$\lambda e^{-\lambda x}$	$x \geq 0$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	Values with a bell-shaped distribution (continuous)	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$	$-\infty < x < \infty$	μ	σ
Standard normal (Z)	Standard scores	$\frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$	$Z = \frac{x-\mu}{\sigma}$	0	1

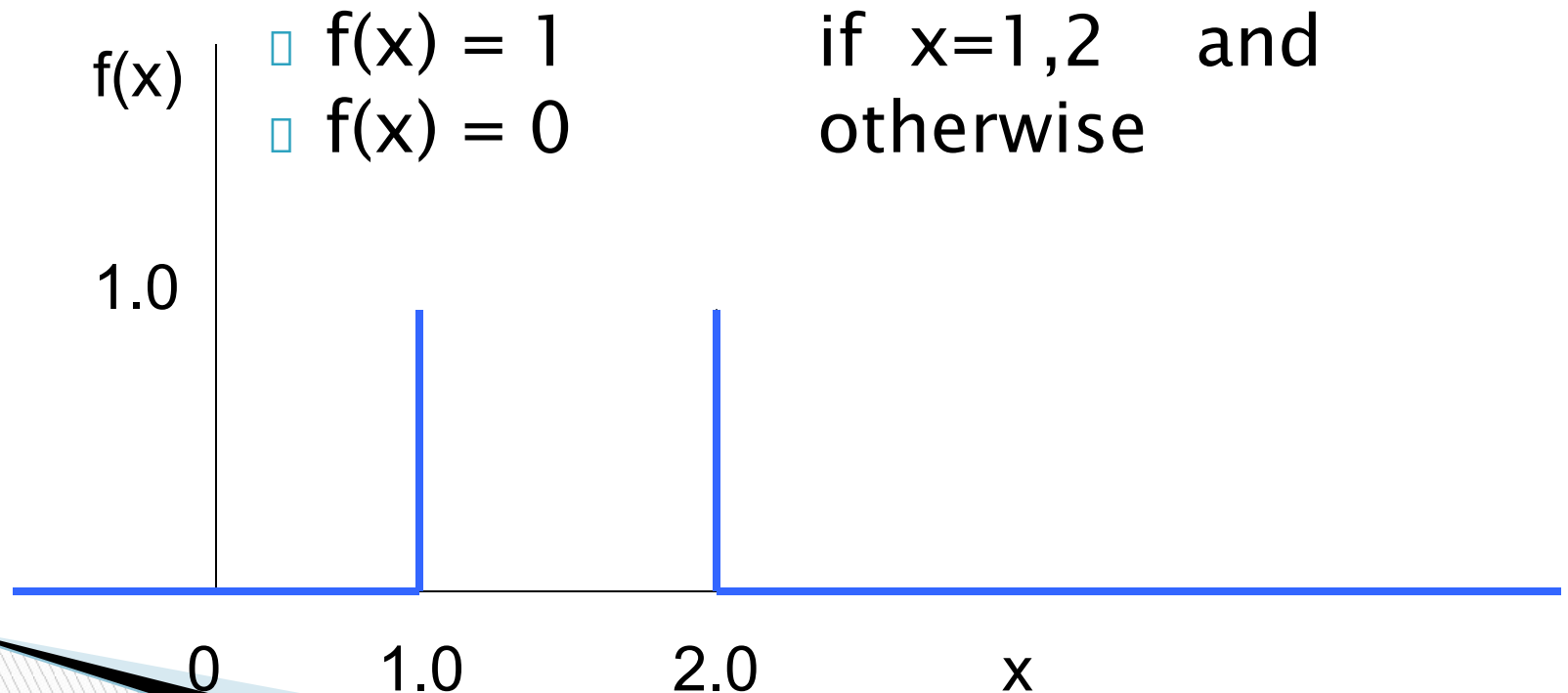
Types of Distributions

☐ Discrete Distribution

- ☐ Uniform
- ☐ Bernoulli
- ☐ Binomial
- ☐ Poisson
- ☐ Geometric

Uniform Distribution

- The **discrete uniform distribution** has the following form:



Bernoulli Distribution

- ▣ An experiment consists of one trial. It can result in one of 2 outcomes: Success or Failure (or a characteristic being Present or Absent).
- ▣ Probability of Success is p ($0 < p < 1$)
- ▣ $Y = 1$ if Success (Characteristic Present), 0 if not

$$p(y) = \begin{cases} p & y = 1 \\ 1 - p & y = 0 \end{cases}$$

$$E(Y) = \sum_{y=0}^1 yp(y) = 0(1-p) + 1p = p$$

$$E(Y^2) = 0^2(1-p) + 1^2p = p$$

$$\Rightarrow V(Y) = E(Y^2) - [E(Y)]^2 = p - p^2 = p(1-p)$$

$$\Rightarrow \sigma = \sqrt{p(1-p)}$$

Reminder:

$$= C_k^n = \frac{n!}{k! (n - k)!}$$

Binomial distribution

A process with two possible outcomes, "success" and "failure" (or yes/no, etc.) is called a *Bernoulli trial*.

e.g. coin tossing: Heads or Tails
 quality control: Satisfactory or Unsatisfactory
 Polling: Agree or disagree

An experiment consists of n independent Bernoulli trials and p = probability of success for each trial. Let X = total number of successes in the n trials.

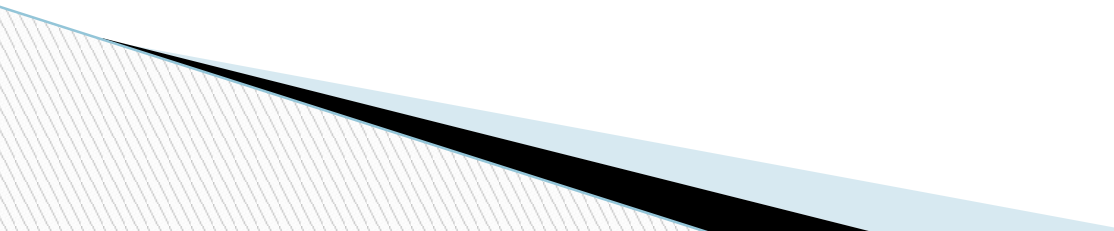
$$\text{Then } P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \text{ for } k = 0, 1, 2, \dots, n.$$

This is called the **Binomial distribution** with parameters n and p , or $B(n, p)$ for short.
 $X \sim B(n, p)$ stands for "X has the Binomial distribution with parameters n and p ."

Binomial Experiment

- ❑ Experiment consists of a series of n identical trials
- ❑ Each trial can end in one of 2 outcomes: Success or Failure
- ❑ Trials are independent (outcome of one has no bearing on outcomes of others)
- ❑ Probability of Success, p , is constant for all trials
- ❑ Random Variable Y , is the number of Successes in the n trials is said to follow Binomial Distribution with parameters n and p
- ❑ Y can take on the values $y=0,1,\dots,n$
- ❑ Notation: $Y \sim \text{Bin}(n,p)$

Situations where a Binomial might occur

- 1) Quality control: select n items at random; X = number found to be satisfactory.
 - 2) Survey of n people about products A and B; X = number preferring A.
 - 3) Telecommunications: n messages; X = number with an invalid address.
 - 4) Number of items with some property above a threshold; e.g. X = number with height $> A$
- 

Binomial distribution

Discrete random variable $X \sim B(n, p)$ if X is the number of successes in n independent Bernoulli trials each with probability p of success

Number k of successes from n independent Bernoulli (YES/NO) trials

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Mean and variance of a binomial distribution

If $X \sim B(n, p)$,

$$\mu = E(X) = np$$

$$\sigma^2 = \text{var}(X) = np(1 - p)$$

Example: A component has a 20% chance of being a faulty. If five are selected from a large batch, what is the probability that more than one is a faulty?

Answer:

Let X = number of faulty in selection of 5

Bernoulli trial: faulty or not faulty, $X \sim B(5, 0.2)$

$$\begin{aligned} P(\text{More than one faulty}) &= P(X > 1) = 1 - P(X \leq 1) = 1 - P(X = 0) - P(X = 1) \\ &= 1 - C_0^5 0.2^0 (1 - 0.2)^5 - C_1^5 0.2^1 (1 - 0.2)^4 \\ &= 1 - 1 \times 1 \times 0.8^5 - 5 \times 0.2 \times 0.8^4 \\ &= 1 - 0.32768 - 0.4096 \approx 0.263. \end{aligned}$$



In the French population about 20% of people prefer Le Pen to other candidates.

An opinion poll asks 1 000 people if they will vote for Le Pen (YES) or not (NO). The expected number of Le Pen voters (YESs) in the poll is therefore $\mu = np = 200$

What is the standard deviation (approximately)?



The number of *YES* votes has a distribution $X \sim B(1000, 0.2)$

The variance is therefore $\sigma^2 = np(1 - p) = 1000 \times 0.2 \times (1 - 0.2)$

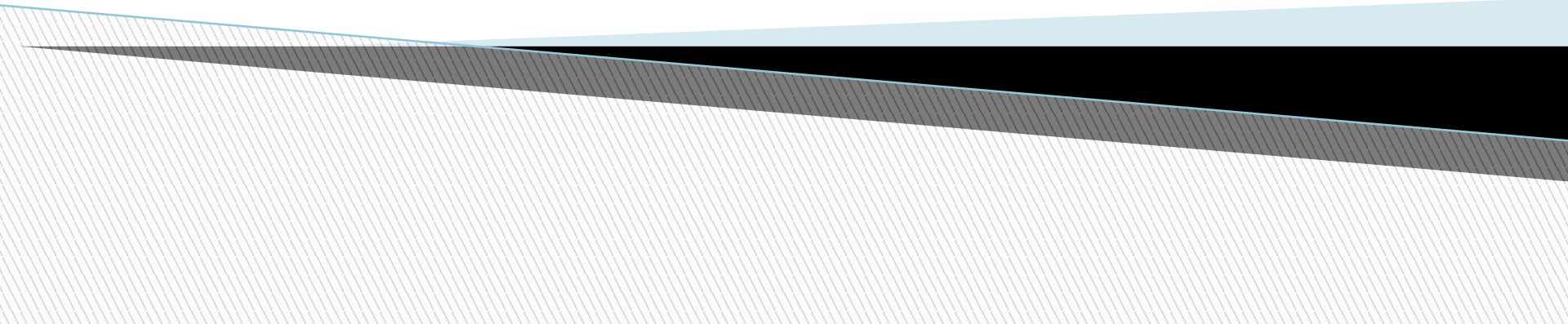
$$= 1000 \times 0.2 \times 0.8 = 160$$

$$\Rightarrow \sigma = \sqrt{\sigma^2} = \sqrt{160} \approx 12.6$$

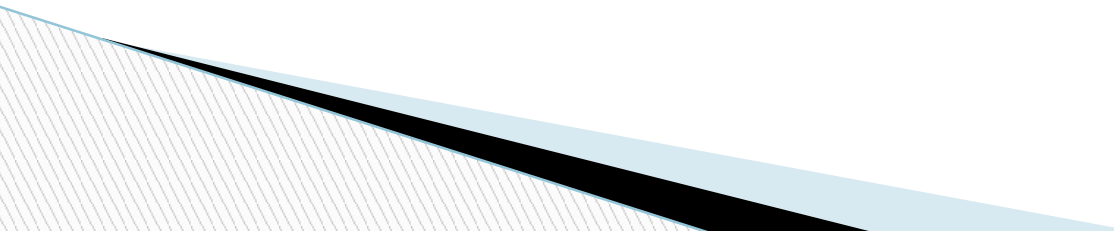
$\Rightarrow$ expect 200 ± 12.6 in poll to say Le Pen

Geometric Distribution

A probability distribution to determine the probability that success will occur on the n th trial of a binomial experiment



Geometric Distribution

- Repeated binomial trials
 - Continue until first success
 - Find probability that first success comes on n th trial
 - Probability of success on each trial = p
- 

Geometric Probability

Probability of success on nth trial =

$$P(n) = p(1 - p)^{n-1}$$

$$\text{mean} = \mu = \frac{1}{p}$$

$$\text{standard deviation} = \sigma = \frac{\sqrt{1 - p}}{p}$$

A sharpshooter normally hits the target 70% of the time.

- Find the probability that her first hit is on the second shot.
- Find the mean and the standard deviation of this geometric distribution.

A sharpshooter normally hits the target 70% of the time.

- **Find the probability that her first hit is on the second shot.**

$$P(2)=p(1-p)^{n-1} = .7(.3)^{2-1} = 0.21$$

- **Find the mean**

$$\mu = 1/p = 1/.7 \approx 1.43$$

- **Find the standard deviation**

$$\sigma = \frac{\sqrt{1-p}}{p} = \frac{\sqrt{1-.7}}{.7} \approx 0.78$$

Poisson distribution

If events happen independently of each other, with average number of events in some fixed interval λ , then the distribution of the number of events k in that interval is **Poisson**.

A random variable X has the Poisson distribution with parameter $\lambda(> 0)$ if

$$P(X = k) = \frac{e^{-\lambda} \lambda^k}{k!} \quad (k = 0, 1, 2, \dots)$$

Examples of possible Poisson distributions

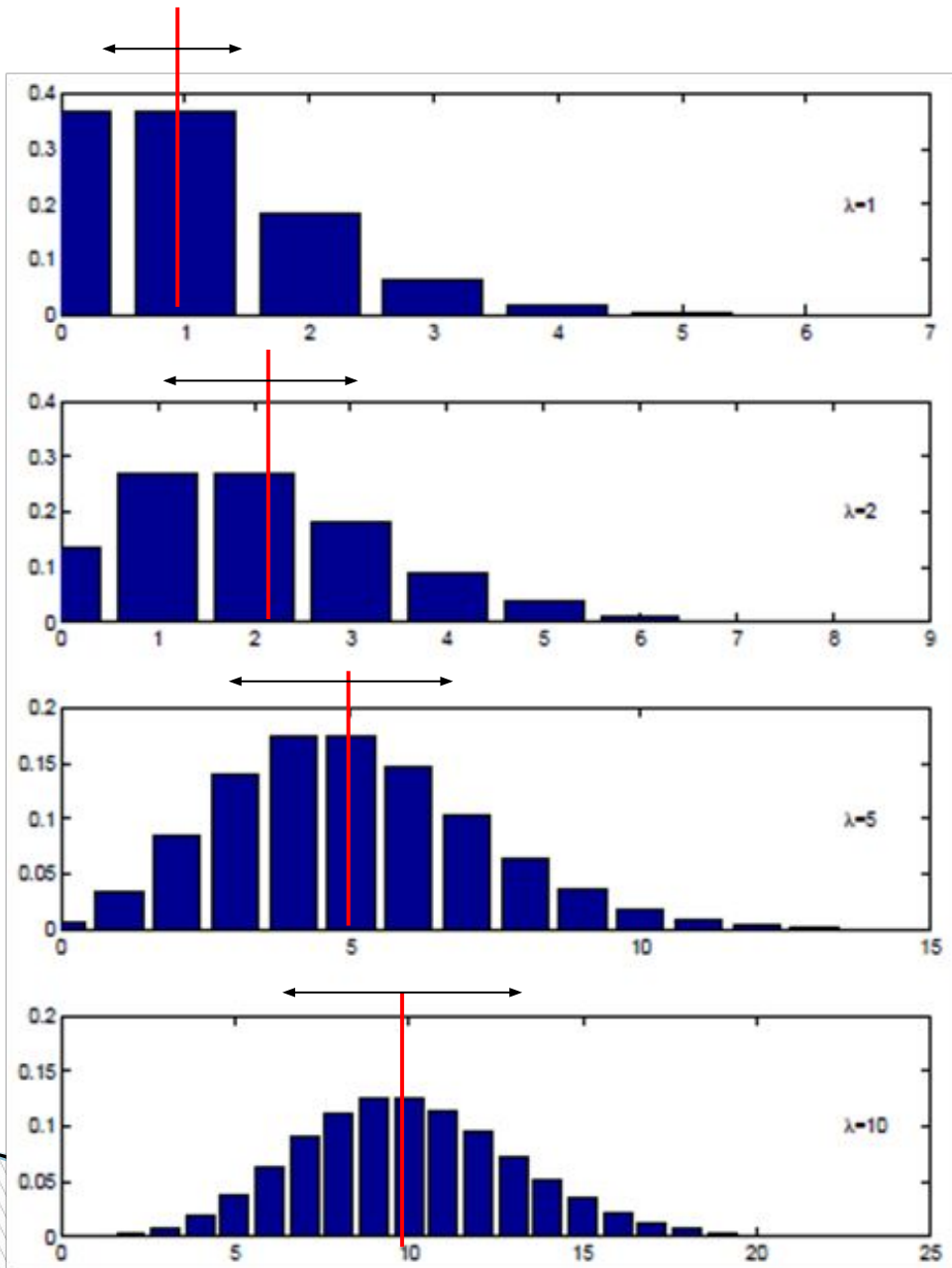
- 1) Number of messages arriving at a telecommunications system in a day
- 2) Number of flaws in a metre of fibre optic cable
- 3) Number of radio-active particles detected in a given time
- 4) Number of photons arriving at a CCD pixel in some exposure time (e.g. astronomy observations)

Mean and variance

If $X \sim \text{Poisson}$ with mean λ , then

$$\mu = E(X) = \lambda$$

$$\sigma^2 = \text{var}(X) = \lambda$$



Example

The probability of a certain part failing within ten years is 10^{-6} . Five million of the parts have been sold so far.

What is the probability that three or more will fail within ten years?

Answer:

Let X = number failing in ten years, out of 5,000,000; $X \sim B(5000000, 10^{-6})$

Evaluating the Binomial probabilities is rather awkward; better to use the Poisson approximation.

X has approximately Poisson distribution with $\lambda = np = 5000000 \times 10^{-6} = 5$.

$$\begin{aligned} P(\text{Three or more fail}) &= P(X \geq 3) = 1 - P(X = 0) - P(X = 1) - P(X = 2) \\ &= 1 - \frac{e^{-5}5^0}{0!} - \frac{e^{-5}5^1}{1!} - \frac{e^{-5}5^2}{2!} \\ &= 1 - e^{-5}(1 + 5 + 12.5) = 0.875 \end{aligned}$$

Example: Telecommunications

Messages arrive at a switching centre at random and at an average rate of 1.2 per second.

(a) Find the probability of 5 messages arriving in a 2-sec interval.

Answer:

Times of arrivals form a Poisson process, rate $\nu = 1.2/\text{sec}$.

(a) Let Y = number of messages arriving in a 2-sec interval.

Then $Y \sim \text{Poisson}$, mean number $\lambda = \nu t = 1.2 \times 2 = 2.4$

$$P(Y = k = 5) = \frac{e^{-\lambda} \lambda^k}{k!} = \frac{e^{-2.4} 2.4^5}{5!} = 0.060$$

Question: (b) For how long can the operation of the centre be interrupted, if the probability of losing one or more messages is to be no more than 0.05?

Answer

(b) Let the required time = t seconds. Average rate of arrival is 1.2/second.

Let k = number of messages in t seconds, so that

$$k \sim \text{Poisson, with } \lambda = 1.2 \times t = 1.2t$$

$$\text{Want } P(\text{At least one message}) = P(k \geq 1) = 1 - P(k = 0) \leq 0.05$$

$$P(k = 0) = \frac{e^{-\lambda} \lambda^k}{k!} = \frac{e^{-1.2t} (1.2t)^0}{0!} = e^{-1.2t}$$

$$\Rightarrow 1 - e^{-1.2t} \leq 0.05$$

$$\Rightarrow -e^{-1.2t} \leq 0.05 - 1$$

$$\Rightarrow e^{-1.2t} \geq 0.95$$

$$\Rightarrow -1.2t \geq \ln(0.95) = -0.05129$$

$$\Rightarrow t \leq 0.043 \text{ seconds}$$

<i>X</i>	<i>X Counts</i>	<i>p(x)</i>	<i>Values of X</i>	<i>E(x)</i>	<i>V(x)</i>
Binomial	Number of successes in n fixed trials	$\binom{n}{x} p^x (1-p)^{n-x}$	$x = 0, 1, \dots, n$	np	$np(1-p)$
Poisson	Number of arrivals in a fixed time period	$\frac{e^{-\lambda} \lambda^x}{x!}$	$x = 0, 1, 2, \dots$	λ	λ
Geometric	Number of trials up through 1st success	$(1-p)^{x-1} p$	$x = 1, 2, 3, \dots$	$\frac{1}{p}$	$\frac{1-p}{p^2}$

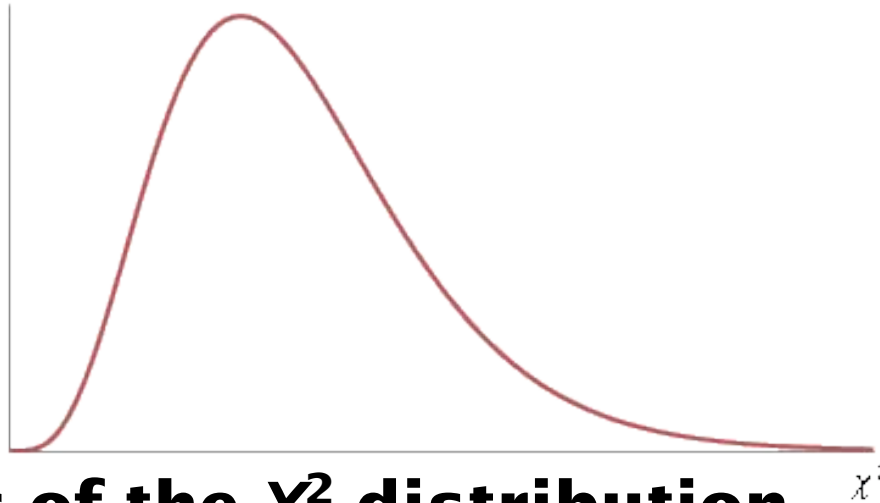
Chi Square distribution

- The Chi Square distribution is the distribution of the sum of squared *standard normal deviates*
- If a simple random sample size n is obtained from a normally distributed population with mean μ and standard deviation σ , then

$$\chi^2 = \frac{(n-1) s^2}{\sigma^2}$$

has a **chi-square distribution** with $n-1$ degrees of freedom.

Chi Square distribution



- ▣ **Properties of the χ^2 distribution**
- ▣ It is *not* symmetric.
- ▣ The shape depends on the degrees of freedom.
- ▣ As the number of degrees of freedom increases, the distribution becomes more symmetric.
- ▣ $\chi^2 \geq 0$

Standard Normal and Student - t

□ Standard Normal z

- $X_1, \dots, X_n$ are independently distributed (μ, σ) , and

$$\bar{X}_n = (X_1 + \dots + X_n)/n$$

$$S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

- then

$$Z = \frac{\bar{X}_n - \mu}{\sigma / \sqrt{n}}$$

is normally distributed
with mean 0 and std
dev 1

□ Student's t-distribution

- Applies to the case where the true standard deviation σ is unknown and is replaced by its sample estimate S_n

$$T = \frac{\bar{X}_n - \mu}{S_n / \sqrt{n}},$$

In the case where X and Y are both discrete random variables whose possible values are, respectively, $x_1, x_2, \dots$, and $y_1, y_2, \dots$, we define the *joint probability mass function* of X and Y , $p(x_i, y_j)$, by

$$p(x_i, y_j) = P\{X = x_i, Y = y_j\}$$

Definition 4.2. A pair (X, Y) of **jointly discrete** random variables has a finite or countably infinite joint range, and the joint distribution is captured by the **joint probability mass function (joint PMF)**, which tells us the individual probability of each pair:

$$P(X = x \cap Y = y), \quad (x, y) \in \text{Range}(X, Y).$$

Given the joint PMF, we can compute the joint distribution by adding:

$$P((X, Y) \in A) = \sum_{(x,y) \in A \cap \text{Range}(X,Y)} \sum P(X = x \cap Y = y), \quad A \subseteq \mathbb{R}^2.$$

This is a consequence of *countable additivity*.

$P(X = x \cap Y = y)$		Y				$P(X = x)$
		1	2	3	4	
X	1	1 / 16	1 / 16	1 / 16	1 / 16	1 / 4
	2	1 / 16	1 / 16	1 / 16	1 / 16	1 / 4
	3	1 / 16	1 / 16	1 / 16	1 / 16	1 / 4
	4	1 / 16	1 / 16	1 / 16	1 / 16	1 / 4
$P(Y = y)$		1 / 4	1 / 4	1 / 4	1 / 4	1

EXAMPLE 4.3a Suppose that 3 batteries are randomly chosen from a group of 3 new, 4 used but still working, and 5 defective batteries. If we let X and Y denote, respectively, the number of new and used but still working batteries that are chosen, then the joint probability mass function of X and Y , $p(i, j) = P\{X = i, Y = j\}$, is given by

$i \backslash j$	0	1	2	3	Row Sum $= P\{X = i\}$
0	$\frac{10}{220}$	$\frac{40}{220}$	$\frac{30}{220}$	$\frac{4}{220}$	$\frac{84}{220}$
1	$\frac{30}{220}$	$\frac{60}{220}$	$\frac{18}{220}$	0	$\frac{108}{220}$
2	$\frac{15}{220}$	$\frac{12}{220}$	0	0	$\frac{27}{220}$
3	$\frac{1}{220}$	0	0	0	$\frac{1}{220}$
Column Sums = $P\{Y = j\}$	$\frac{56}{220}$	$\frac{112}{220}$	$\frac{48}{220}$	$\frac{4}{220}$	

Definition 5.1

Two random variables X and Y are **jointly continuous** if there exists a nonnegative function $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}$, such that, for any set $A \subset \mathbb{R}^2$, we have

$$P((X, Y) \in A) = \iint_A f_{XY}(x, y) dx dy \quad (5.15)$$

The function $f_{XY}(x, y)$ is called the **joint probability density function (PDF)** of X and Y .

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{XY}(x, y) dx dy = 1$$

Marginal PDFs

$$f_X(x) = \int_{-\infty}^{\infty} f_{XY}(x, y) dy, \quad \text{for all } x,$$
$$f_Y(y) = \int_{-\infty}^{\infty} f_{XY}(x, y) dx, \quad \text{for all } y.$$

The joint density function of X and Y is given by

$$f(x, y) = \begin{cases} 2e^{-x}e^{-2y} & 0 < x < \infty, 0 < y < \infty \\ 0 & \text{otherwise} \end{cases}$$

Compute **(a)** $P\{X > 1, Y < 1\}$; **(b)** $P\{X < Y\}$;

$$P\{X > 1, Y < 1\} = \int_0^1 \int_1^\infty 2e^{-x}e^{-2y} dx dy$$

$$\begin{aligned} P\{X > 1, Y < 1\} &= \int_0^1 \int_1^\infty 2e^{-x} e^{-2y} dx dy \\ &= \int_0^1 2e^{-2y} (-e^{-x} |_1^\infty) dy \\ &= e^{-1} \int_0^1 2e^{-2y} dy \\ &= e^{-1} (1 - e^{-2}) \end{aligned}$$

$$\begin{aligned} P\{X < Y\} &= \iint_{(x,y):x<y} 2e^{-x}e^{-2y} \, dx \, dy \\ &= \int_0^\infty \int_0^y 2e^{-x}e^{-2y} \, dx \, dy \\ &= \int_0^\infty 2e^{-2y}(1 - e^{-y}) \, dy \\ &= \int_0^\infty 2e^{-2y} \, dy - \int_0^\infty 2e^{-3y} \, dy \\ &= 1 - \frac{2}{3} \\ &= \frac{1}{3} \end{aligned}$$

Consider two events A and B with nonzero probabilities, $P(A)$ and $P(B)$. The **conditional probability** of event A given event B is

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

The conditional probability of event B given event A is

$$P(B|A) = \frac{P(A \cap B)}{P(A)}$$

The **probability of the intersection** of two events A and B is

$$\begin{aligned} P(A \cap B) &= P(A)P(B|A) \\ &= P(B)P(A|B) \end{aligned}$$

A corporation is proposing to select two of its current regional managers as vice presidents. In the history of the company, there has never been a female vice president. The corporation has six male regional managers and four female regional managers. Make the assumption that the 10 regional managers are equally qualified and hence all possible groups of two managers should have the same chance of being selected as the vice presidents. Now find the probability that both vice presidents are male.

Solution Let A be the event that the first vice president selected is male and let B be the event that the second vice president selected is also male. The event that represents both selected vice presidents are male is the event (A and B)—that is, the event $A \cap B$. Therefore, we want to calculate $P(A \cap B) = P(B|A)P(A)$, using Definition 4.8.

For this example,

$$P(A) = P(\text{first selection is male}) = \frac{\text{\# of male managers}}{\text{\# of managers}} = \frac{6}{10}$$

$$\begin{aligned}
 P(B|A) &= P(\text{second selection is male given first selection was male}) \\
 &= \frac{\text{\# of male managers after one male manager was selected}}{\text{\# of managers after one male manager was selected}} \\
 &= \frac{5}{9}
 \end{aligned}$$

Thus,

$$P(A \cap B) = P(A)P(B|A)$$

$$\frac{6}{10} \left(\frac{5}{9} \right) = \frac{30}{90} = \frac{1}{3}$$

Two events A and B are **independent events** if

$$P(A|B) = P(A) \quad \text{or} \quad P(B|A) = P(B)$$

(*Note:* You can show that if $P(A|B) = P(A)$, then $P(B|A) = P(B)$, and vice versa.)

Definition 4.9 leads to a special case of $P(A \cap B)$. When events A and B are independent, it follows that

$$P(A \cap B) = P(A)P(B|A) = P(A)P(B)$$

Bayes Theorem Formula

For any two events A and B, Bayes's formula for the Bayes theorem is given by:

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Where,

- **P(A)** and **P(B)** are the probabilities of events A and B; also, P(B) is never equal to zero.
- **P(A|B)** is the probability of event A when event B happens,
- **P(B|A)** is the probability of event B when A happens.

Bayes Theorem Statement

Bayes' Theorem for n sets of events is defined as,

Let $E_1, E_2, \dots, E_n$ be a set of events associated with the sample space S , in which all the events $E_1, E_2, \dots, E_n$ have a non-zero probability of occurrence. All the events $E_1, E_2, \dots, E$ form a partition of S . Let A be an event in space S for which we have to find the probability, then according to Bayes theorem,

$$P(E_i | A) = \frac{P(E_i) \cdot P(A|E_i)}{\sum_{k=1}^n P(E_k) \cdot P(A|E_k)}$$

for $k = 1, 2, 3, \dots, n$

Bayes Theorem Derivation

The proof of Bayes' Theorem is given as, according to the conditional probability formula,

$$P(E_i | A) = \frac{P(E_i \cap A)}{P(A)} \dots\dots \text{(i)}$$

Then, by using the multiplication rule of probability, we get

$$P(E_i \cap A) = P(E_i) \cdot P(A | E_i) \dots\dots \text{(ii)}$$

Now, by the total probability theorem,

$$P(A) = \sum_{k=1}^n P(E_k) \cdot P(A | E_k) \dots\dots \text{(iii)}$$

Substituting the value of $P(E_i \cap A)$ and $P(A)$ from eq (ii) and eq(iii) in eq(i) we get,

$$P(E_i | A) = \frac{P(E_i) \cdot P(A | E_i)}{\sum_{k=1}^n P(E_k) \cdot P(A | E_k)}$$